

Association-Rule Mining with Apriori on Online Retail: A Browser-Based Reproducibility Study

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Assignment: Association Rules (Week 4)

Abstract

Online shops need "frequently bought together" suggestions, but raw co-occurrence counts promote items that are merely popular. We built a browser-only Apriori miner over 17,080 UCI Online Retail baskets and ranked rules by support, confidence, and lift at support $\geq 1\%$ and confidence $\geq 30\%$. It produced 950 rules, all with lift above 1, including a china-pattern pair with confidence 78.29% and lift 17.1440. Confidence alone misleads whenever the consequent is popular, so lift — not confidence — should decide what ships.

Index Terms — association rules, Apriori, lift, market-basket analysis

1. Introduction

Problem statement. Mine "baskets with A also contain B" rules at support, confidence, and lift thresholds, keeping only lift > 1 . Confidence ignores consequent frequency, so rules into popular items pass any fixed floor almost automatically. Rule counts explode non-linearly — halving support multiplies rules more than tenfold — and co-occurrence never implies causation.

Motivation. Fourth assignment of an AI-assisted Recommender Systems course, moving from Week 3 personalised similarity to the co-occurrence signal. Association rules power "frequently bought together" widgets with no user model: the non-personalised baseline every shop needs before personalisation.

Concrete example. With defaults (support 1.0%, confidence 30%), Run rules returns 950 rules; clicking GREEN $\rightarrow$ ROSES teacup shows 78.29% confidence at lift 17.1440, and Reverse direction gives 69.36% at identical lift.

Contributions.

- Browser Apriori app implementing the seven TODO(hw4) stubs (baskets, counts, three metrics, miner, bidirectional rules) with a sortable results table; no runtime framework.
- Verified numbers — 11/11 harness, a hand-checked rule, a 1,200-row independent cross-check with 0 mismatches — plus the required useful/misleading, threshold, direction, and validation analyses.

2. Related Work

Prior work consulted: the UCI Online Retail data and case study [1] [2], the Apriori papers [3] [4], and association-rule background [5].

Alternatives considered. *FP-growth* — faster at 0.5% support, rejected: the brief requires Apriori and posting-list intersection is already interactive. *Confidence-only ranking* — rejected: Results (B) clears 30% confidence while sitting at the lift floor. *Week-3 collaborative filtering* — needs user history; rules

serve anonymous shoppers.

3. Method

Approach. Baskets are InvoiceNo groups from UCI Online Retail (id 352, Dec 2010–Dec 2011) [2] after the starter cleaning (no cancellations, positive quantities and prices, no guests or non-products): 17,080 baskets, 3,653 items, no per-basket timestamps. dedupeBasket collapses repeats in first-appearance order so support means share of baskets; metrics are $\text{support} = \text{count}(A \cup B)/N$, $\text{confidence} = \text{count}(A \cup B)/\text{count}(A)$, $\text{lift} = \text{confidence}/(\text{count}(B)/N)$, each returning {value, defined} with defined=false on zero denominators. Top item 85123A (WHITE HANGING HEART) sits in 1,959 baskets (11.47%); nothing exceeds ~11.5%.

Pipeline. (1) dedupeBasket builds the basket index. (2) countItemset intersects sorted posting lists. (3) findFrequentItemsets runs level-wise Apriori, pruning candidates with any infrequent (k-1)-subset [3] [4]. (4) generateRules splits each itemset both ways (bitmask over non-empty proper subsets), filters by confidence, and lifts against global N. (5) The page applies the slider thresholds, renders the sortable table, and opens a detail panel with Reverse direction per rule.

Tools & libraries.

- HTML5 / CSS3 / vanilla JavaScript (ES6), no runtime framework — docs: <https://developer.mozilla.org/>
- Python 3.11.9, local cross-check script only (not shipped) — <https://www.python.org/>
- Microsoft Edge (Chromium) headless --print-to-pdf for the submitted PDF — <https://www.microsoft.com/edge>
- Git / GitHub CLI for version control and Pages deploy — <https://cli.github.com/>

Key design decisions. *Sets, not multisets:* a quantity-30 line counts once, matching "share of baskets". *Both directions with global-N lift:* confidences differ while lifts are provably identical. *Thresholds 1%/30% + lift > 1:* 1% (≈ 171 baskets) admits whole product families while 30% cuts one-off co-occurrences; lift > 1 is automatic here (see §4) but guards lower floors.

Configuration. Support slider 0.5–10% (default 1.0%), confidence slider 10–100% (default 30%); deterministic code, no random seeds; runs in any modern browser; repo PollyIva/Association-Rules-Recommender, branch main.

4. Experiments

Setup. Final table at 1%/30% with lift > 1 retained. Extra runs at 0.5%/30%, 3%/30%, 1%/60% (plus 1%/90% and 3%/60% for the sweep). Full exports saved for the 950-, 12-, and 238-row settings; the 10,723-row 0.5%/30% run is cited by status-line count. Screenshots of both analysed rules are saved in useful rule/ and weak rule/.

Worked example (assignment §6.7 hand-check): 23171 $\rightarrow$ 23172 with counts 269/224/202 and N = 17,080 gives support $202/17080 = 1.18\%$, confidence $202/269 = 75.09\%$, lift $0.750929/(224/17080) = 57.2584$ — exactly what the page shows; reversed confidence is $202/224 = 90.18\%$ at identical lift.

Results (A): useful rule (spec §4.2 item 1). GREEN $\rightarrow$ ROSES REGENCY TEACUP: 541/691/780, support 3.17%, confidence 78.29%, lift **17.1440** (reverse 69.36%). Same-pattern pair in 1 of 31 baskets — ship as a "Complete the set" slot plus bundle. The ROSES base rate is only 4.57%, so buyers are $\sim 17\times$ enriched: lift, not confidence, makes the case.

Controls

Minimum support 1.0%

0.5% – 10% of baskets.

Minimum confidence 30%

10% – 100%.

Run rules Run tests

Done — 950 rule(s) at support $\geq$ 1.0% and confidence $\geq$ 30%.

Test log

Press “Run tests” to execute the self-checks.

Rules

950 rules.

Antecedent (A)	Consequent (B)	count(AuB)	count(A)	count(B)	support	confidence	lift
22386 — JUMBO BAG PINK POLKADOT	850998 — JUMBO BAG RED RETROSPOT	546	869	1584	3.20%	62.83%	6.7749
850998 — JUMBO BAG RED RETROSPOT	22386 — JUMBO BAG PINK POLKADOT	546	1584	869	3.20%	34.47%	6.7749
22697 — GREEN REGENCY TEACUP AND SAUCER	22699 — ROSES REGENCY TEACUP AND SAUCER	541	691	780	3.17%	78.29%	17.1440
22699 — ROSES REGENCY TEACUP AND SAUCER	22697 — GREEN REGENCY TEACUP AND SAUCER	541	780	691	3.17%	69.36%	17.1440
22726 — ALARM CLOCK BAKELIKE	22727 — ALARM CLOCK BAKELIKE	530	784	877	3.10%	67.60%	13.1658

Fig. — 1%/30% run: teacup rule in the results table

Selected rule

Antecedent (A) 22699 — ROSES REGENCY TEACUP AND SAUCER

Consequent (B) 22697 — GREEN REGENCY TEACUP AND SAUCER

count(AuB) 541

count(A) 780

count(B) 691

support 3.17%

confidence 69.36%

lift 17.1440

Reverse direction (B → A): confidence **78.29%**, lift **17.1440**. Confidence changes with direction; lift does not.

Reverse direction (B → A)

Fig. — Teacup rule detail, reversed: 69.36% confidence, lift 17.1440

Results (B): misleading rule (spec §4.2 item 2). HOME BUILDING BLOCK WORD → 85123A: 209/690/1959, support 1.22%, confidence 30.29%, lift **2.6409** (second-lowest; minimum 2.6366). Into 85123A, lift = confidence/0.1147, so any rule at confidence $\geq$ 30% scores $\geq$ 2.62: this rule sits at the mathematical floor and inherits the bar from the consequent's popularity.

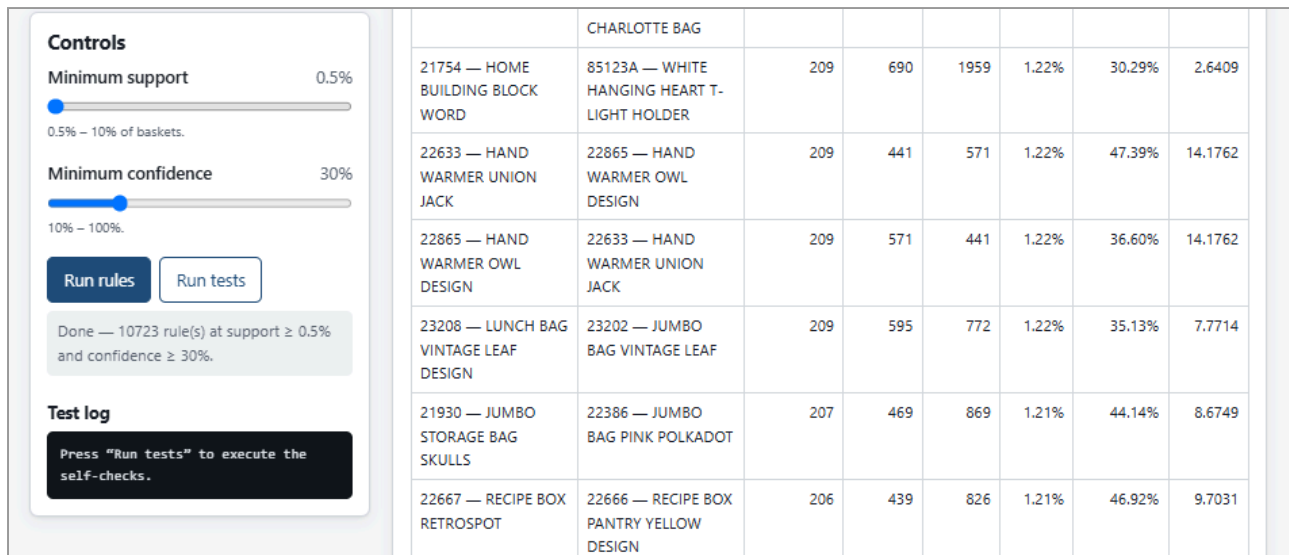


Fig. — 0.5%/30% run: weak rule in the results table

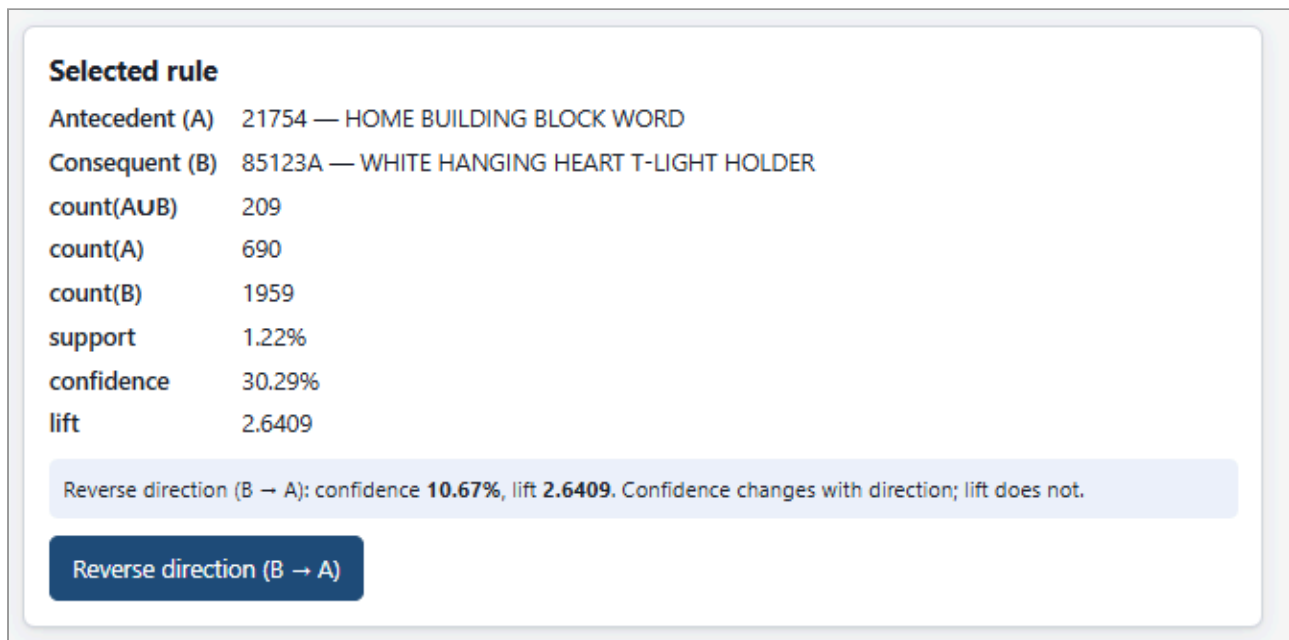


Fig. — Weak rule detail: 209/690/1959, 30.29% confidence, lift 2.6409

Results (C): thresholds (spec §4.2 item 3).

Support	Confidence	Itemsets	Rules
0.5%	30%	5,255	10,723
1%	30%	1,219	950
3%	30%	115	12
1%	60%	1,219	238
1%	90%	1,219	18
3%	60%	115	5

Support acts before generation — downward closure compounds (1%→0.5% unlocks pairs and triples up to size 6) — while confidence only filters afterwards: itemsets stay 1,219 as rules fall 950→238→18.

Direction (optional). Confidences differ whenever counts differ (teacups $78.29 \leftrightarrow 69.36$; tea plates $75.09 \leftrightarrow 90.18$; jumbo bags $62.83 \leftrightarrow 34.47$) while lifts stay identical (17.1440, 57.2584, 6.7749). Deploy the direction whose page view is the trigger; lift only says the pair is worth deploying.

Comparison vs. baseline. The baseline is independence (lift = 1): at 1%/30% every rule clears it by arithmetic (lift $\geq 0.30/0.1147 \approx 2.62$), so the lift filter removes nothing. It matters at lower floors: at confidence $\geq 10\%$ the page admits CAKESTAND $\rightarrow$ WHITE HEART at 13.92% confidence with lift 1.2137 — barely above chance — and an offline 0.5%/0.5% re-mine found 12 of 21,026 rules with lift < 1 (minimum 0.9267).

Verification. Harness 11/11 (baseline was 2 passed / 9 pending); hand-check matches in both directions; all 1,200 exported rows re-derived from raw counts with 0 mismatches and exact header counts; exports plus four screenshots saved.

5. Discussion

Failure case. A draft called the BLOCK-WORD rule (lift 2.6409) the table minimum; the exhaustive cross-check sort printed a lower row first:

```
23321 SMALL WHITE HEART OF WICKER  $\rightarrow$  85123A ... lift 2.6366
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Root cause. The claim came from eyeballing rows near the table bottom instead of sorting all 950 — the bottom region is dominated by rules into 85123A without being ordered among themselves.

Fix + verification. *Detection:* the sorted cross-check output above. *Change:* corrected to second-lowest with both numbers stated. *Confirmation:* re-sorted the full export — the five lowest lifts all point at 85123A and all 950 exceed 1.

What worked.

- Posting-list Apriori: 5,255 itemsets mined interactively with no libraries.
- Lift-first presentation with base rates shown, so "30% confident" versus "17 $\times$ enriched" is visible on the page.
- Export-based evidence: every number traceable to a file, which is what caught the \$5 error above.

What surprised me.

- The required lift > 1 filter is mathematically redundant at my own operating point (floor 2.62).
- A 2 $\times$ threshold loosening multiplies rules more than 10 $\times$ — the sliders act at different pipeline stages with different units of effect.

Next improvement. Collect timestamped baskets, mine on months 1–7, and keep only rules whose months 8–9 lift stays above 1 at no less than half the period-1 support; then A/B-test the teacup bundle on attach rate before shipping.

6. AI Usage Disclosure

AI tools used. OpenCode (AI coding CLI, model big-pickle): code generation, debugging, cross-check assembly, documentation.

What I personally verified. Ran the page (11/11 tests, all cited settings, both rules, reverse button, sortable headers); re-checked every number against the exports and re-did the hand-check arithmetic; chose the examples and confirmed the lowest-lift ranks by sorting the full export.

What I trusted without verification. The starter export's cleaning (assignment §3) — the raw CSV was never shipped, so baskets were taken as given; slider-to-fraction wiring by code reading plus spot checks; screenshots as representative of each setting rather than re-captured per run.

References

- [1] D. Chen, S. L. Sain, and K. Guo, "Data mining for the online retail industry: A case study of RFM model-based customer segmentation using data mining," *J. Database Mark. Cust. Strateg. Manag.*, vol. 19, no. 3, 2012. (As listed on the live UCI page [2]; the assignment sheet lists *J. Cases Inf. Technol.* — we follow the verifiable source.)
- [2] D. Chen, "Online Retail [Dataset]," UCI Machine Learning Repository, 2015. [Online]. Available: <https://archive.ics.uci.edu/dataset/352/online+retail>. [Accessed: 2026-10-08].
- [3] R. Agrawal, T. Imieliński, and A. Swami, "Mining association rules between sets of items in large databases," in *Proc. ACM SIGMOD Int. Conf. Manage. Data*, Washington, DC, USA, 1993, pp. 207–216. [Online]. Available: <https://doi.org/10.1145/170035.170072>. [Accessed: 2026-10-08].
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- [5] Wikipedia contributors, "Association rule learning," Wikipedia. [Online]. Available: https://en.wikipedia.org/wiki/Association_rule_learning. [Accessed: 2026-10-08].